Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Switching and VLANS

Task:

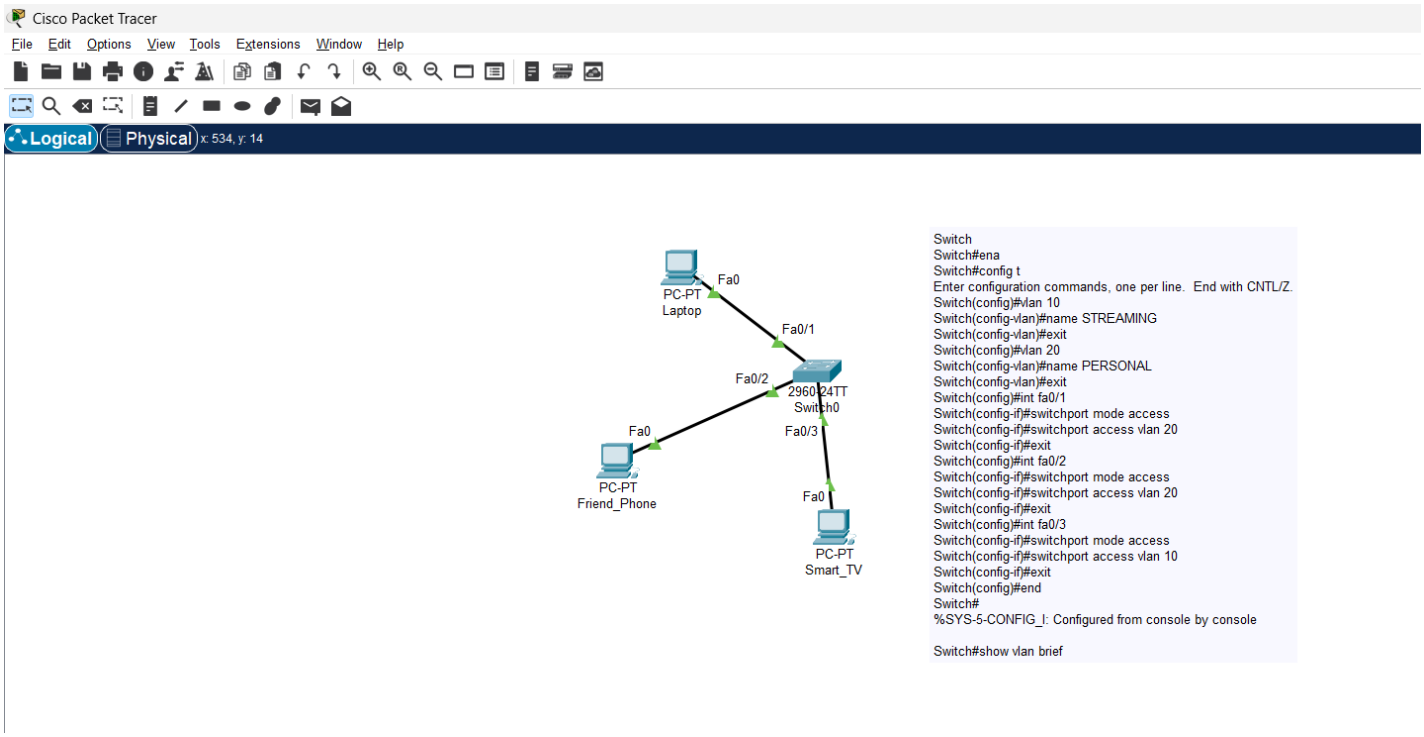
- **1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).**
  - **Aim**
    - To understand network switching and Virtual Local Area Networks (VLANs) by designing a simple network diagram in which three devices are connected to a switch and assigned to different VLANs.

  - **Introduction**
    - **Switching** is a networking process in which a network switch forwards data frames to the appropriate device using MAC addresses. A switch connects multiple devices within a Local Area Network (LAN) and helps them communicate efficiently.
    - A **VLAN (Virtual Local Area Network)** is a logical division of a physical network into separate broadcast domains. VLANs allow devices connected to the same physical switch to be grouped into different logical networks.

- VLANs improve network organization, reduce unnecessary broadcast traffic between groups, and help separate devices according to their purpose.
- In this assignment, three devices—a laptop, a friend's phone, and a smart TV—are connected to a switch. VLAN 10 is used for streaming devices, while VLAN 20 is used for personal devices.

➢ **Network Diagram**

➢ Draw the following diagram in your assignment:



➢ **Diagram explanation:** All three devices connect to the same physical switch. The laptop and friend's phone belong to VLAN 20, while the smart TV belongs to VLAN 10.

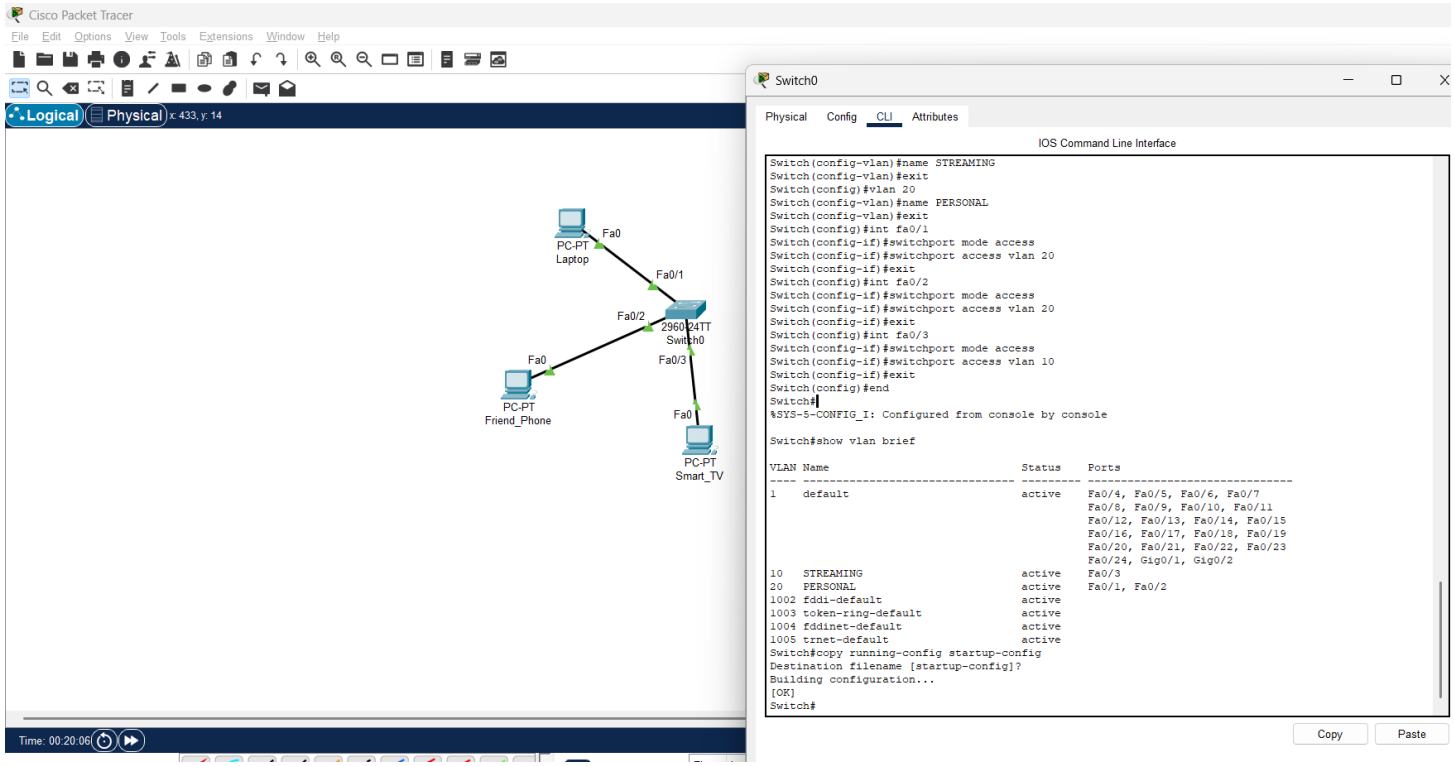
- **VLAN Assignment Table**

Sr. No.	Device	Purpose	VLAN ID	VLAN Name
1	Laptop	Personal use	20	PERSONAL
2	Friend's Phone	Personal use	20	PERSONAL
3	Smart TV	Video streaming	10	STREAMING

- **Explanation of VLANs**
- **VLAN 10 – STREAMING**
- VLAN 10 is created for streaming devices such as smart TVs. It separates streaming traffic from the personal-device group at Layer 2.
- **VLAN 20 – PERSONAL**
- VLAN 20 is created for personal devices such as laptops and phones. It groups personal devices into a separate logical network.
- **Role of the Switch**
- The switch connects all three devices and forwards Ethernet frames according to its MAC address table and VLAN configuration. Each switch port is assigned to the appropriate VLAN.
- By default, devices in different VLANs cannot communicate directly at Layer 2. Communication between VLANs requires a router or a Layer 3 switch with suitable configuration.

- **Advantages of VLANs**
  1. **Network Segmentation:** VLANs divide a physical network into logical groups.
  2. **Reduced Broadcast Traffic:** Broadcast traffic is generally limited to devices within the same VLAN.
  3. **Improved Organization:** Devices can be grouped according to their purpose.
  4. **Better Access Control:** VLAN separation can support security policies when combined with suitable network controls.
  5. **Efficient Network Management:** Administrators can manage different device groups more easily.

- **Result**
- A simple network diagram was prepared showing a laptop, a friend's phone, and a smart TV connected to a switch. The laptop and phone were assigned to VLAN 20 (PERSONAL), while the smart TV was assigned to VLAN 10 (STREAMING).


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 433, y: 14
PC-PT Laptop
Fa0
Fa0/1
Fa0/2
2960-24TT Switch0
Fa0/3
Fa0
PC-PT Friend_Phone
Fa0
PC-PT Smart_TV
Time: 00:20:06
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch(config-vlan)#name STREAMING
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name PERSONAL
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/4, Fa0/5, Fa0/6, Fa0/7
Fa0/8, Fa0/9, Fa0/10, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1, Gig0/2
10 STREAMING active Fa0/3
20 PERSONAL active Fa0/1, Fa0/2
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
Copy
Paste

❖ 2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

➢ Aim

- To configure a simulated network switch in Cisco Packet Tracer by creating two VLANs, assigning three PCs to VLAN 10 and two PCs to VLAN 20, and verifying that PCs belonging to different VLANs cannot communicate directly using the ping command.

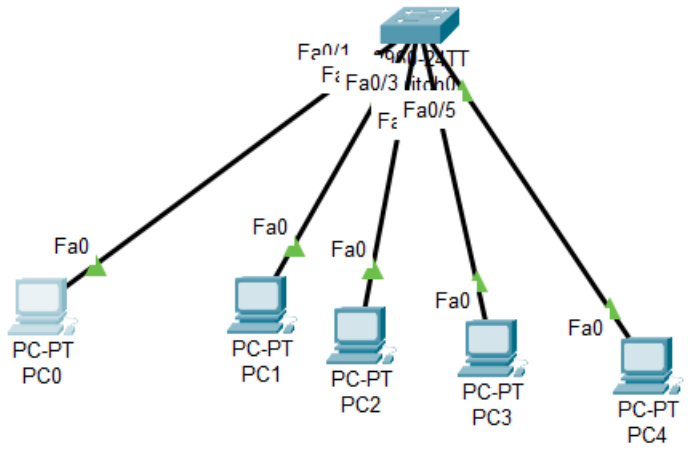
➢ Introduction

- A **VLAN (Virtual Local Area Network)** is a logical division of a physical network into separate broadcast domains. VLANs allow a network administrator to group devices according to their purpose, department, or security requirements.
- A switch can support multiple VLANs. Each switch port can be assigned to a specific VLAN using configuration commands. Devices connected to access ports in the same VLAN can communicate at Layer 2, provided their IP configuration is correct.
- Devices belonging to different VLANs are separated into different broadcast domains. They cannot communicate directly through a Layer 2 switch unless inter-VLAN routing is configured using a router or a Layer 3 switch.
- In this practical, VLAN 10 and VLAN 20 are configured on a Cisco switch. Three PCs are assigned to VLAN 10, and two PCs are assigned

to VLAN 20. Ping tests are performed to verify connectivity within and between the VLANs.

- **Tools and Requirements**
  - Cisco Packet Tracer software
  - One Cisco 2960 switch
  - Five PC-PT devices
  - Copper Straight-Through cables
  - Switch CLI and PC Command Prompt

- **4. Network Topology**
- The network consists of one switch and five PCs.

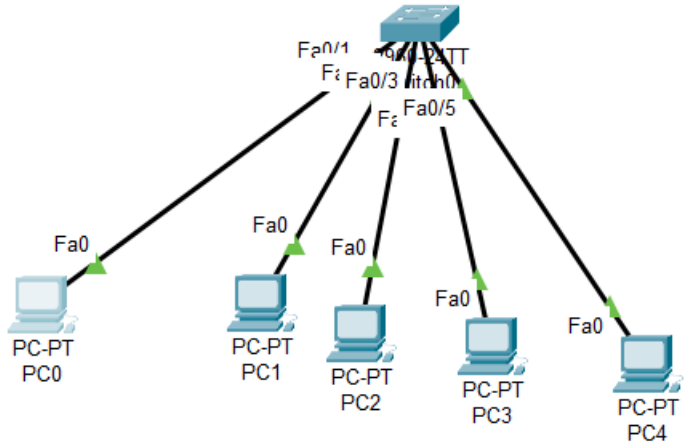


- **5VLAN and IP Address Assignment**

Device	Switch Port	VLAN ID	IP Address	Subnet Mask
PC0	Fa0/1	10	192.168.10.2	255.255.255.0
PC1	Fa0/2	10	192.168.10.3	255.255.255.0
PC2	Fa0/3	10	192.168.10.4	255.255.255.0
PC3	Fa0/4	20	192.168.20.2	255.255.255.0
PC4	Fa0/5	20	192.168.20.3	255.255.255.0

- For this practical, PCs in different VLANs use different IP subnets. No router or Layer 3 switch is configured for inter-VLAN routing.

- **Procedure**
- **Step 1: Create the Network**
  1. Open Cisco Packet Tracer.
  2. Place one Cisco 2960 switch in the workspace.
  3. Add five PC-PT devices.
  4. Connect each PC to a separate switch port using Copper Straight-Through cables.

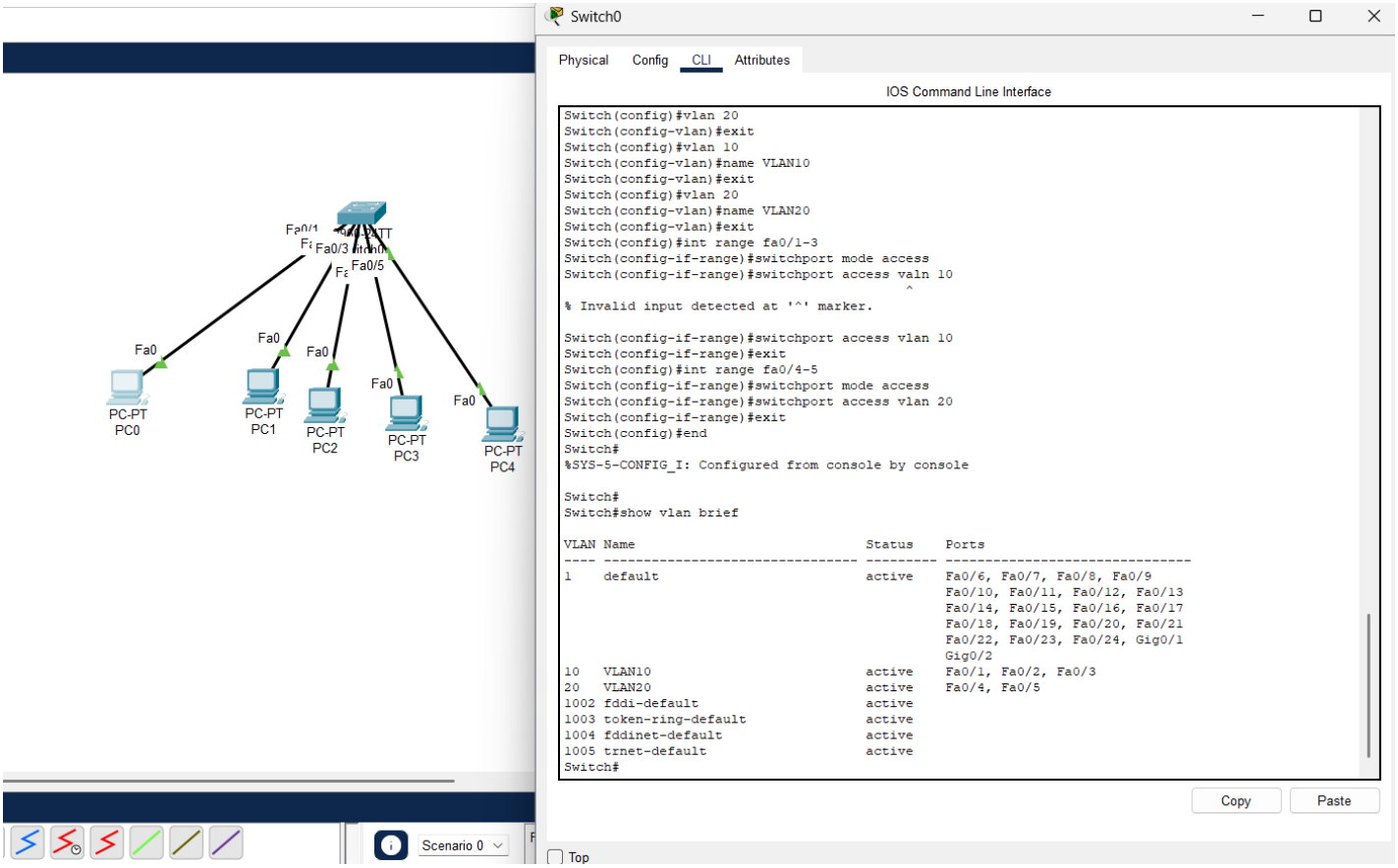


- **Step 2: Create VLAN 10 and VLAN 20**
- Open the switch CLI and enter the following commands:

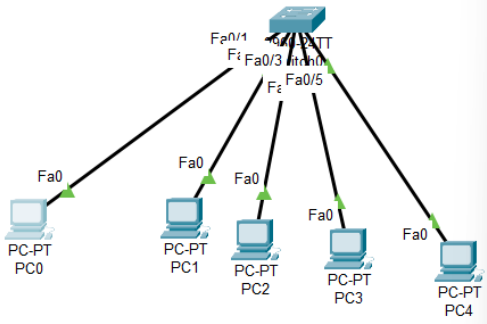
  enable
  configure terminal
  vlan 10
  name VLAN10
  exit
  vlan 20
  name VLAN20
  exit
- These commands create VLAN 10 and VLAN 20 on the switch.
- **Step 3: Assign Three PCs to VLAN 10**
- Enter the following commands:

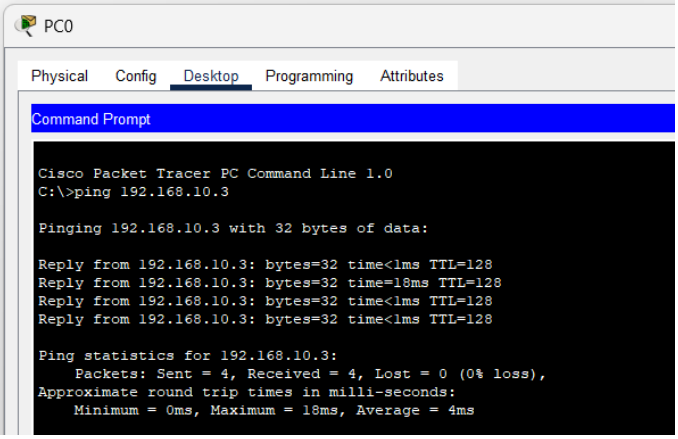
interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit

- These commands assign switch ports Fa0/1, Fa0/2, and Fa0/3 to VLAN 10.
- **Step 4: Assign Two PCs to VLAN 20**
- Enter the following commands:
  interface range fa0/4-5
  switchport mode access
  switchport access vlan 20
  exit
  end
- These commands assign switch ports Fa0/4 and Fa0/5 to VLAN 20.
- **Step 5: Configure the PC IP Addresses**
- For each PC, open **Desktop → IP Configuration**, select **Static**, and enter the IP address and subnet mask from the table in Section 5.
- No default gateway is required for this demonstration because inter-VLAN routing is not configured.
- **Step 6: Verify the VLAN Configuration**
- In the switch CLI, enter:
  show vlan brief
- The output should show:
  - VLAN 10 containing ports Fa0/1, Fa0/2, and Fa0/3.
  - VLAN 20 containing ports Fa0/4 and Fa0/5.

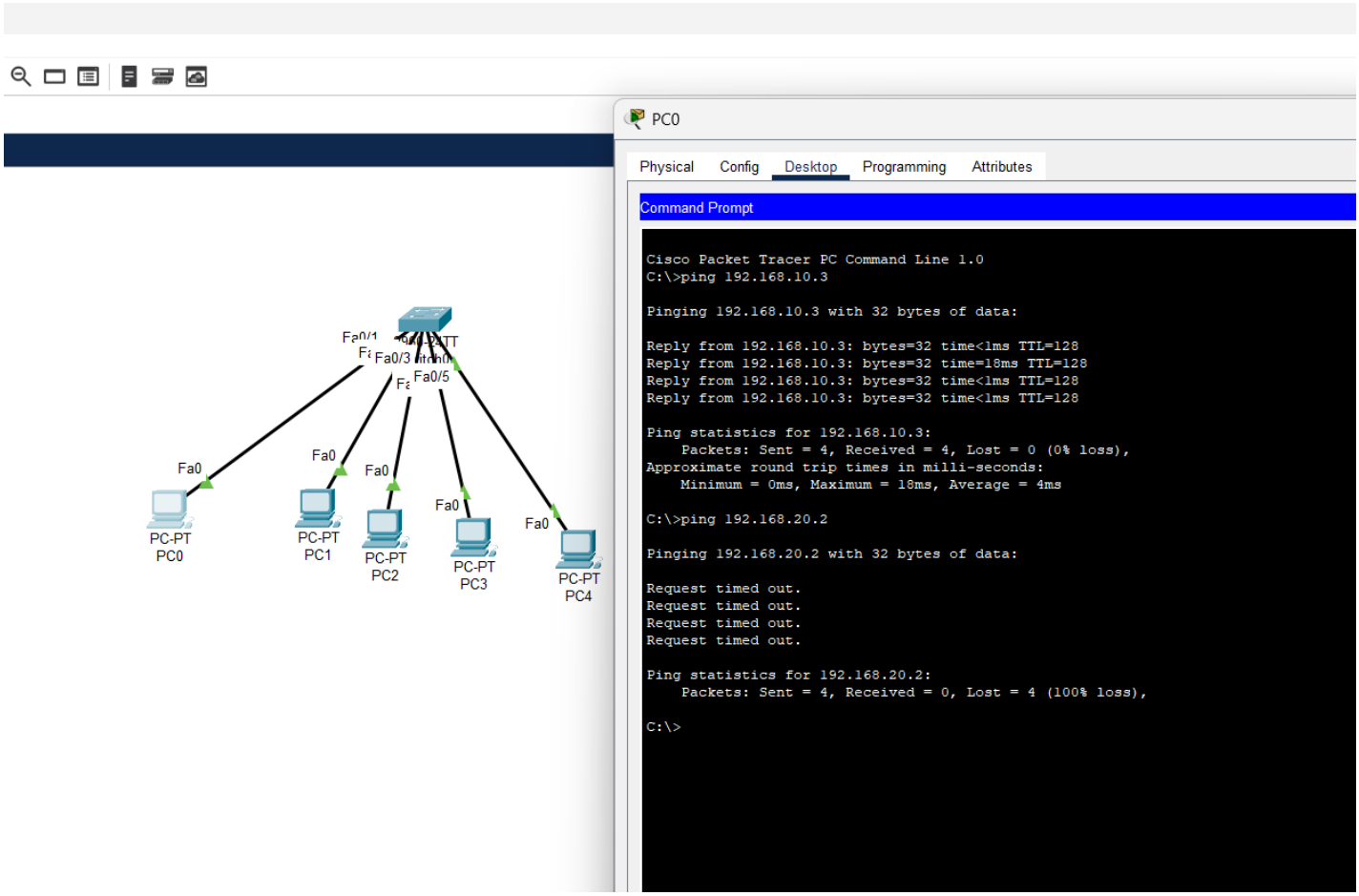


- **Step 7: Test Connectivity Within VLAN 10**
- On PC0, open **Desktop → Command Prompt** and enter:
  ping 192.168.10.3
- A successful ping indicates that PC0 can communicate with PC1 in the same VLAN.





- **Step 8: Test Connectivity Between Different VLANs**
- On PC0, enter:
  ping 192.168.20.2
- This tests connectivity from PC0 in VLAN 10 to PC3 in VLAN 20. The ping should fail because inter-VLAN routing has not been configured.



➢ Observation Table

Test	Source	Destination	Expected Result
1	PC0 (VLAN 10)	PC1 (VLAN 10)	Successful ping

2	PC1 (VLAN 10)	PC2 (VLAN 10)	Successful ping
3	PC3 (VLAN 20)	PC4 (VLAN 20)	Successful ping
4	PC0 (VLAN 10)	PC3 (VLAN 20)	Ping fails
5	PC2 (VLAN 10)	PC4 (VLAN 20)	Ping fails

- The table shows the expected results. Actual observations should be recorded from the Packet Tracer simulation.

- **Result**
- Two VLANs were created on the Cisco switch. Three PCs were assigned to VLAN 10, and two PCs were assigned to VLAN 20. Connectivity tests demonstrated that PCs within the same VLAN could communicate, while PCs in different VLANs could not communicate directly because inter-VLAN routing was not configured.

❖ 3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

➢ Aim

- To move a PC from VLAN 10 to VLAN 20 in a network simulator, verify the switch configuration before and after the change, and test network connectivity using the ping command.

➢ Introduction

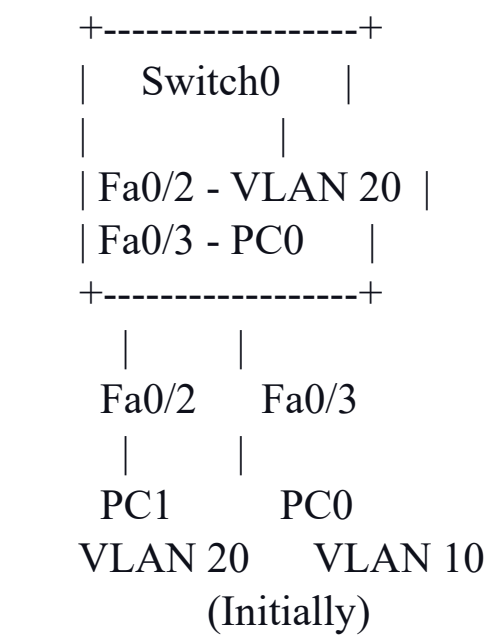
- A **VLAN (Virtual Local Area Network)** is used to divide a physical network into separate logical networks. Each VLAN creates a separate broadcast domain and helps improve network organization, security, and traffic management.
- In a switched network, a PC can be moved from one VLAN to another by changing the VLAN assignment of the switch port to which the PC is connected. The physical connection does not need to be changed.
- In this practical, a PC connected to switch port **Fa0/3** is initially assigned to VLAN 10. The port is then reassigned to VLAN 20. The configuration is verified using show vlan brief, and connectivity is tested using the ping command.

➢ Tools and Requirements

- Cisco Packet Tracer or another network simulator
- One Cisco 2960 switch
- Two PCs
- Copper Straight-Through cables

- Switch CLI and PC Command Prompt

➢ **4. Network Topology**



➢ **IP Address Configuration**

Device	Switch Port	Initial VLAN	IP Address	Subnet Mask
PC0	Fa0/3	VLAN 10	192.168.20.2	255.255.255.0
PC1	Fa0/2	VLAN 20	192.168.20.3	255.255.255.0

- For this practical, both PCs use the same IP subnet so that the effect of changing the VLAN assignment can be demonstrated clearly. No router or inter-VLAN routing is required.
- **Procedure and Configuration Commands**
- **Step 1: Create VLAN 10 and VLAN 20**
- Open the switch CLI and enter:

```
enable
configure terminal

vlan 10
name VLAN10
exit

vlan 20
name VLAN20
exit
```

- These commands create VLAN 10 and VLAN 20 on the switch.
- **Step 2: Before Configuration — Assign PC0 to VLAN 10**
- Configure the switch ports as follows:

```
interface fa0/3
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
```

exit
end

- To verify the initial configuration, enter:
  show vlan brief
- **Expected observation before the change:**
- VLAN 10 contains port Fa0/3, connected to PC0.
- VLAN 20 contains port Fa0/2, connected to PC1.
- Therefore, PC0 is initially a member of VLAN 10.
- **Step 3: Test Connectivity Before the Change**
- On PC0, open **Desktop → Command Prompt** and enter:
  ping 192.168.20.3
- **Expected result:** The ping fails because PC0 belongs to VLAN 10, while PC1 belongs to VLAN 20. Inter-VLAN routing has not been configured.
- **Step 4: After Configuration — Move PC0 to VLAN 20**
- To move PC0 from VLAN 10 to VLAN 20, enter these commands on the switch:
  enable
  configure terminal

  interface fa0/3
  switchport mode access
  switchport access vlan 20
  exit

  end
- The command switchport access vlan 20 changes the access VLAN of port Fa0/3 from VLAN 10 to VLAN 20.

- The cable and physical port remain unchanged.
- **Step 5: Verify the New VLAN Configuration**
- Enter the following command:
  show vlan brief
- **Expected observation after the change:**
- VLAN 10 no longer contains port Fa0/3.
- VLAN 20 contains ports Fa0/2 and Fa0/3.
- This confirms that PC0 has been moved to VLAN 20.
- **Step 6: Test Connectivity After the Change**
- On PC0, open **Desktop → Command Prompt** and run:
  ping 192.168.20.3
- **Expected result:** The ping succeeds because PC0 and PC1 now belong to VLAN 20 and are in the same IP subnet.
- If the first ping times out, run the command again because the devices may need to resolve each other's MAC addresses.

- **Before and After Comparison**

Parameter	Before Change	After Change
PC0 switch port	Fa0/3	Fa0/3
PC0 VLAN assignment	VLAN 10	VLAN 20
PC1 VLAN assignment	VLAN 20	VLAN 20
show vlan brief	Fa0/3 appears under VLAN 10	Fa0/3 appears under VLAN 20

Ping from PC0 to PC1	Fails	Succeeds

- **Result**
- PC0 was successfully moved from VLAN 10 to VLAN 20 by changing the VLAN assignment of switch port Fa0/3. The show vlan brief command verified the change. Connectivity testing demonstrated that communication with PC1 failed before the change and succeeded after both PCs were placed in VLAN 20, assuming the IP configuration was correct.

❖ **4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.**

- **Aim**
  - To automate the assignment of 10 switch ports to VLAN 30 using Cisco IOS commands in Cisco Packet Tracer.

- **Introduction**
  - A **VLAN (Virtual Local Area Network)** is used to logically divide a network into separate broadcast domains. A switch can assign multiple ports to the same VLAN to organize network devices and manage communication.

- In Cisco Packet Tracer, the interface range command allows a network administrator to configure multiple switch ports simultaneously instead of entering the same command separately for each port.
- In this practical, VLAN 30 is created, and 10 switch ports are assigned to it using a single range command.

➢ **Tools and Requirements**
➢ Cisco Packet Tracer
➢ Cisco 2960 Switch
➢ Switch CLI

➢ **Configuration Commands**
➢ The following commands are used to create VLAN 30 and assign the first 10 Fast Ethernet ports to it.

```
enable
configure terminal

vlan 30
name VLAN30
exit

interface range fa0/1-10
switchport mode access
switchport access vlan 30
exit

end
```

- **Explanation of Commands**

  enable: Enters privileged EXEC mode.
  configure terminal: Enters global configuration mode.
  vlan 30: Creates VLAN 30 on the switch.
  name VLAN30: Assigns a name to VLAN 30.
  interface range fa0/1-10: Selects switch ports Fa0/1 through Fa0/10 simultaneously.
  switchport mode access: Configures the selected ports as access ports.
  switchport access vlan 30: Assigns all selected ports to VLAN 30.
  exit: Exits the interface configuration.
  end: Returns to privileged EXEC mode.

- **Verification**
- To verify that all 10 ports have been assigned to VLAN 30, enter:
  show vlan brief
- The output should show VLAN 30 containing the following ports:
- Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5,
- Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10
- This confirms that all 10 selected switch ports belong to VLAN 30.

- **Result**
- VLAN 30 was successfully created, and 10 switch ports were assigned to it using the interface range command. The configuration was verified using the show vlan brief command.